

Investigating Sensory Perception as a Material for Fashion Design

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ABSTRACT

The increasing digitalization of fashion design opens up new potentials for designing and experiencing fashion. The study presented in this paper aimed to investigate new potentials for artistic expression by deconstructing the human body into materials for design using technological devices. The senses of sight, hearing, and touch were used as materials for design by creating a distance between these stimuli and the body's surface and relocating them to the surface of a bodysuit that was designed for this purpose. Undergraduate students participated in the study by exploring bodily perception from the perspective of fashion design in a workshop. The findings of the workshop suggest potentials for fashion designers to use technologies as design tools for designing dress beyond the textile surface by turning the human body, in terms of bodily senses, into material for design.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods;

KEYWORDS

Body as material, Fashion technology, Future fashion, Perception as material, Sensing dress

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1 INTRODUCTION

Body and dress are commonly perceived simultaneously in the moment in which dress is being worn. From the external perspective, dress serves as a visual extension of the body that not only alters the visual silhouette but positions the body in a social context and thus makes it socially acceptable through conformity [6]. From the internal perspective dress can expand bodily perception in various ways. Layers of dress can extend one's haptic perception when they are touched [20] and influence whether one feels warm or cold [10], or how one moves one's body [27].

In fashion design processes, however, the body and dress are considered to be two separate entities. This stands in contrast to the aforementioned dynamics between the body and dress in daily life. Designers commonly design dress for the body without using the body as a material for design [1, 15]. While traditional design practices in fashion such as draping, which is the practice most strongly related to the body, use the body as a three-dimensional reference to create dress on its surface, pattern-construction techniques, which are predominant in the fashion industry, make little use of the body and only consider normative body measurement when developing different kinds of dress [1].

This tendency to disregard the body as a material for fashion design is further emphasized in the current trend of digitalizing dress-making processes. Digital computer-aided design (CAD) prototyping software such as Clo3D and Brozwear allows designers to program digital representations of the human body and textile materials to act and react similarly to their physical counterparts [11, 30]. This makes it possible to design dress digitally from and for the external perspective, resulting in a more sustainable and efficient development process that reduces resource consumption and the need for physical prototyping [3, 9, 14, 18]. However, the current use of technology in fashion completely disregards the internal perspective on designing dress, which is crucial with regard to everyday experiences of body-dress expressions. Thus, the fashion industry demonstrates its resistance to a more holistic change by only using digital tools to make use of centuries-old methods and processes rather than explore the artistic potential of designing dress in new and innovative ways [1, 26].

In the context of fashion, then, the human body is limited in its potential and serves solely as reference material, rather than as a material for design. This contrasts with media industries such as the film- and video-game industries, in which the human body is explored and used creatively in design processes. Here, digital bodies have the same features and characteristics as discussed above, but are not regarded solely as reference material and instead also serve as materials for design. Computer-generated imagery (CGI) movies such as *Shrek* and *Frozen* show human-related body forms that deviate in terms of form and texture from what one is used to in relation to the representation of the human body. The possibility of creating a digital body that deviates from its physical counterpart is rooted in the idea that there is no need for the precise replication of a physical body in the digital reality, in terms of its somatic and kinaesthetic approaches to making the digital body function in its context. [12]. In the video-game industry, moreover, digital bodies are not only seen from the outside but controlled from the inside: Digital cameras serve as 'digital eyes' to see on the computer screen, digital microphones serve as 'digital ears' to hear through the computer speakers, and the keyboard and mouse of the

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computer initiate movements of the digital body in the digital space [16]. This demonstrates that (i) the digital body can be reshaped as a material for design to the designer's liking, and (ii) that the designers in this field also design the ways in which digital space is perceived by the digital body.

Fashion designers can take inspiration from the ways in which media industries use digital representations of the human body as materials for design by using technology as a design tool to enrich and broaden their understanding of their design practice. Deconstructing the human body into its somaesthetic and kinaesthetic components through the use of technology can open up new ways of designing dress, both from and for an external and from and for an internal perspective. Thus, this article proposes that the body and dress can be considered to be materials of equal importance in fashion design processes; that they can go beyond the surface of the body by using bodily features, such as bodily perception, as materials for design that are utilized alongside textile materials. Fashion design could also consider how one perceives both one's body and the environment through dress in new ways by including technology as a design tool in the design process, in order to enhance artistic expression.

This article presents the findings of an exploratory workshop that was conducted with fashion design students; the aim was to explore new methods of including perception-altering technology in design processes by using technology to design dress. In this workshop, the temporal union of body and dress, conceived of in terms of body-ownership illusions, was explored using a bodysuit that combined textiles and technologies to alter bodily perception. The exploration was guided by a theoretical framework for embodied design ideation methods, and later analyzed in relation to the participants' described experiences. The analysis of these experiences suggested that the use of technology to alter bodily perception created an awareness when designing dress that considers both the external and internal perspectives when wearing dress.

2 RELATED STUDIES

The altering of the body's senses has been investigated from different perspectives. Some work has focused on the brain's processes of sensory input in relation to the feeling of body ownership, while other studies have focused on augmentation of the body through technology, or the development of textile wearables to increase body awareness in everyday life, to name a few.

2.1 Body Ownership Illusion

Various phenomena have been investigated within the field of neuroscience and psychology in relation to how multisensory processing operated by the brain leads to a sense of body ownership [13]. Studies with a focus on body-ownership illusions have investigated the effects of perceiving non-bodily objects as parts of one's own body by locating bodily sensations such as touch on non-body objects. A number of studies have investigated these effects in terms of 'rubber-hand-illusions'. Botvinick and Cohen created the first reported rubber-hand illusion during a study in which participants experienced rubber hands as their own by placing a rubber hand next to that of the participant, which was concealed from the participant's view. The rubber hand was then touched at the same time

and in a similar manner as the concealed hand, which caused a perceptive drift in which the participant identified the rubber hand as a part of their body [4]. Other studies have suggested that the rubber-hand illusion can be expanded to encompass the full body: in one, participants saw the body of a life-sized mannequin from its own, first-person, perspective by wearing a head-mounted display that was connected to a camera. The participants simultaneously experienced haptic stimulation on their own bodies and the visual input of the mannequin being touched in the same places from the mannequin's first-person perspective. This caused a drift in the participants' body perception in relation to the mannequin's body [19]. Full-body rubber-hand illusions have also been investigated using virtual-reality technologies: In one study, participants experienced a virtual body as their own body by feeling haptic sensations at the moment at which a virtual body was being touched in the same places [7].

2.2 Re-Mixed Bodies

Studies of human-computer interactions have investigated body-ownership illusions with a focus on technological possibilities. *Met-aLimbs* is a pair of wearable robotic arms that was experienced by participants as an integral part of their body; these were controlled by the movement of the legs, allowing participants to control four hands simultaneously. The robotic arms also provided haptic feedback when they touched objects or surfaces, which was experienced by the feet [21]. The *FingerReader* assisted visually impaired people in reading printed texts. The device consisted of a camera that scanned text, which was then read out to the participant using computer software [22]. 'Stimulator suits' as part of the *CyberSM* installation, transmitted haptic sensations from one user to another. Visitors to the installation were able to share touch sensations by using a computer interface to initiate vibration, pressure, or pain remotely [24]. Another installation allowed visitors to explore the mediation of the seeing, hearing and touch through a stationary tele-robot which could rotate its head in response to the participant's head movement. This allowed users to experience the environment through the robot, and to experience their own body from a third-person perspective [25].

2.3 Wearables for Alternating Bodily Perception

In the context of textiles, several studies have explored the altering of the senses through wearables and how these can be used to improve bodily awareness. *In Touch with the Now* presented a jacket that simulate the sensation of being touched on the back when the sleeves are interacted with [5]. *Wear.mascha* was developed as a wearable, and consists of a headpiece and a vest; it is intended to recreate the experience of experiencing a migraine and tinnitus [2]. *Trailblazer* was developed as a wearable navigation tool that also makes the wearer aware of poor posture. Vibration motors located on different parts of the vest create haptic sensations, guiding the wearer in the right direction without the need for visual navigation tools [28]. *Wearable X* is a pair of smart yoga leggings with sensors and haptic motors woven into the fabric at the hips, knees, and ankles to improve how the wearer adopts yoga positions. The sensors measure the distances between each of them, and vibrate

to help the wearer to correct their form [8]. *AWElectric* is a textile wearable that simulates the sensation of goosebumps by vibrating at different points on the wearer's body. This intimate haptic sensation is complemented by visual feedback in terms of changing lights [17].

These studies show that altering bodily perception has been investigated within various disciplines with focus on how one experiences and interacts with one's body as well as with the environment. Investigating the altering of bodily perception from the perspective of fashion design could contribute to this discourse by exploring how the concept can be used from an aesthetic point of view in relation to expressions of the body and dress.

3 WORKSHOP

3.1 Theoretical Framework

A workshop with fashion design undergraduate students as participants was organized to test the concept of bodily perception as material for fashion design. The setup of the workshop was informed by the framework on embodied design ideation (EDI) proposed by Wilde *et. al.* in 'Embodied Design Ideation Methods: analyzing the power of estrangement' [29]. This was undertaken in order to turn tacit knowledge - the participants' experiences - into explicit knowledge that could be interpreted in order to contribute to the findings of this study. For this reason, the four-step structure of 'disruption -> destabilization -> emergence -> embodiment' suggested by Wilde *et. al.* was applied.

The intention was for estrangement as a disruptive element to occur on a physical level by bringing about a perceptive drift in terms of the senses of sight, hearing, and feeling away from the surface of the body towards the outermost layer of the bodysuit. This was intended to cause the destabilization of participants' body-dress distinction by creating body-ownership illusions. This drift in perception was intended to support the emergence of a novel understanding of the body and textiles as materials of equal importance in fashion design processes, and to embody technologically produced, temporal body-ownership experiences.

3.2 Subjects

Four undergraduate fashion design students, ranging in age between 22 and 25 years, were recruited for the workshop; three were female and one was male. The selection of the participants was undertaken based on their area of study. The number of participants was limited to four as the workshop focused strongly on the bodily perception and the experience of the participants. This more personal approach was intended to help the participants to open up more freely and fully engage with one another.

3.3 The Bodysuit

A bodysuit consisting of textile materials and technological devices was developed. This served as a wearable design tool that destabilized perceived body-dress distinctions through body-ownership illusions, created by the disruptive altering of the perception of sight, sound, and touch.

The bodysuit consisted of two layers; the inner layer was worn directly on the skin, while the outer layer was positioned at a distance from the body and thus physically extended it. This was

made possible by attaching a structure to the participants' shoulders, to which the outer layer was attached.

The altering of the aforementioned senses was made possible by embedding or attaching devices to the two layers of the bodysuit. The vision of the wearer was altered using a GoPro Camera Series 7, an iPhone 7, the GoPro Smartphone app, and a head-mounted display made of cardboard. Hearing was altered using a Boya BY-WM4 Mark II sound-recording system, Apple earphones, and Beats headphones with noise canceling function activated. Touch was altered using two Arduino Mega boards, 18 vibration motors, and conductive yarns as capacitive sensors.

3.4 Procedure

The workshop was conducted on October 5, 2020 between 1 pm and 9 pm and was organized in two main sections. During that period, a space of 48m² was accessible only to the participants and the author.

In the first section of the workshop, the participants were asked to experiment with altering one of their senses by attaching one of the devices to their bodies. This was undertaken in order to reduce the likelihood of the participants being overwhelmed by too many novel stimuli when wearing the bodysuit for the first time.

In the second section of the workshop, the participants wore the bodysuit to experience complete altering of the aforementioned senses. The participants were encouraged to walk around and experience the bodysuit and its altering of sight, sound, and touch. The other participants were encouraged to work with sight, sound, and touch in relation to the person wearing the bodysuit. This was repeated until all of the participants had worn the bodysuit.

3.5 Data Analysis

The participants' comments were collected using video recorded during the workshop and written feedback at the end. During the workshop, the participants were asked to reflect on their experiences in terms of disruptions to their senses, destabilized body-dress distinctions, and what emerged as a result of this. A similar structure was used for the written feedback. Each participant was asked the same questions, and to reflect on their experiences during the workshop and report them in written form. The questions focused on the disruption of the body's senses, the destabilization of body-dress distinctions, and the potential emergence of a new understanding of body and dress as materials of equal importance in fashion design processes.

The analysis of the findings was undertaken by comparing the participants' comments to identify similarities and differences. The focus of these comparisons was (i) the participants' understanding of how technology can be used in fashion design, and (ii) how technology can change the body-dress distinction that is prevalent in current fashion design.

4 FINDINGS

The findings are presented in three sub-sections, which focus on disruption, destabilization, and emergence. This structure is intended to support the reader's understanding of how the disruption of sensory perception caused the destabilization of the body-dress distinction, which led to the emergence of the idea of the body and

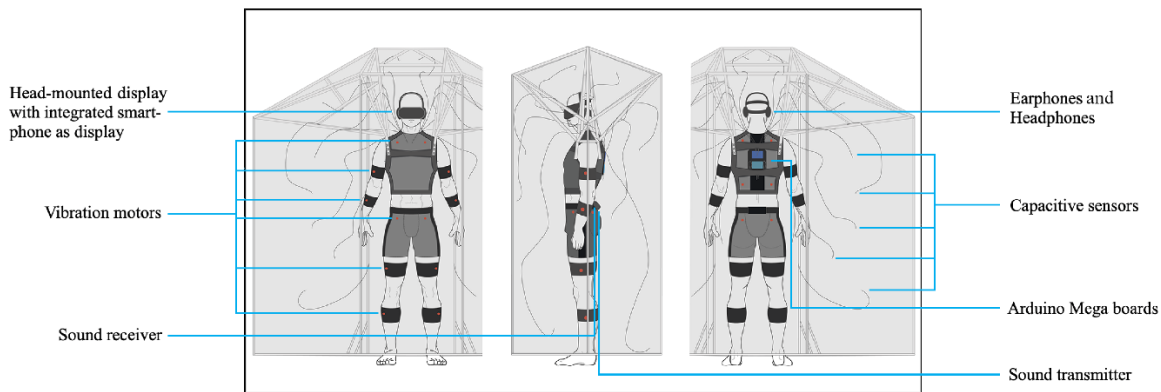


Figure 1: A technical illustration of the bodysuit showing where the different technological devices were placed on the bodysuit.

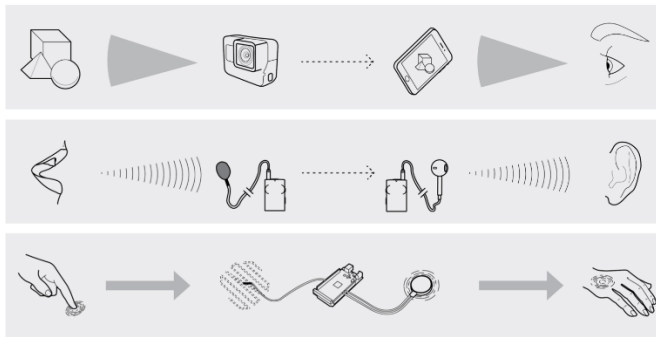


Figure 2: Shows the technical illustration for altering the bodily perception through technological devices. The first row shows the alteration of sight, the second row shows the alteration of hearing, and the third row shows the alteration of touch.

textiles as being materials of equal importance in fashion design processes.

4.1 Drifting Bodily Perception of Sight, Sound, and Touch

The participants' bodily perception of sight, sound, and touch was explored using the technological devices connected to the bodysuit. The portable camera transmitted video material in real-time to the smartphone, which was attached to the inside of the head-mounted display. This setup allowed the sense of sight being altered by placing the camera on different parts of the participants' bodies so that the participants saw the environment from different perspectives: from foot or hand level, or from the back when the camera was placed on the back of the body. The participants described this experience as feeling very small, very big, or as if their bodies were twisted, depending on where the camera was positioned.

A similar approach was taken to explore the altering of hearing; a sound recording system that transmitted the recorded sounds in real-time was connected to earphones worn by a participant. The microphones and earphones altered how the participants perceived

sounds and located the source of the sounds that they were exposed to. The participants felt that footsteps were louder than speech when the microphone was attached to a shoe, and that sounds coming from the back seemed to be closer than sounds from the front when the microphone was attached to one of their backs. The effect was that the participants raised or lowered their voices depending on how far the microphone was from their ears.

The altering of touch was explored by connecting touch sensors and vibration motors, which functioned as 'touch simulators', in different patterns. The effect was that the participants were surprised when they saw the hand of one of the other participants touching the sensors that were closest to the arms of the participant who was wearing the bodysuit, which activated vibration motors positioned on e.g. the back or the legs. The participants felt this experience made them not trust their own senses, as they could not predict in advance where they would feel a sensation based on the visual information they were receiving.

While the altering of the perception of sight, sound, and touch was perceived by all of the participants, the intensity varied for each. The changes were felt by three of the four participants to be most pronounced for sight and touch. Sight was felt to be the most drastic change in terms of how the participants experienced their environment, while the altering of touch was felt to be the most emotionally impactful. The altering of hearing, in comparison, was described as less impactful, which was linked to the white-noise generated by the sound-recording system.

A greater degree of commonality was expressed by the participants in terms of how the altering of the perception of sight, sound, and touch affected their bodily awareness. None of the participants moved initially while wearing the bodysuit, and all struggled to verbalize their thoughts when first wearing it; all stood still for a while and spread their fingers while holding their hands away from their bodies. One participant in particular lost bodily awareness to such a degree that she repeatedly tilted so much to one side that she would have fallen over if the author had not reacted and supported her. The participants eventually adjusted to the new experience, and slowly began to explore their environment while adjusting to the altering of their senses. The initial adjustment phase was described as follows: "The experience put my senses in another



Figure 3: A photograph of the components of the bodysuit. (1) Shows the components for altering sight, (2) show the components for altering sound and (3) shows the components for altering touch.

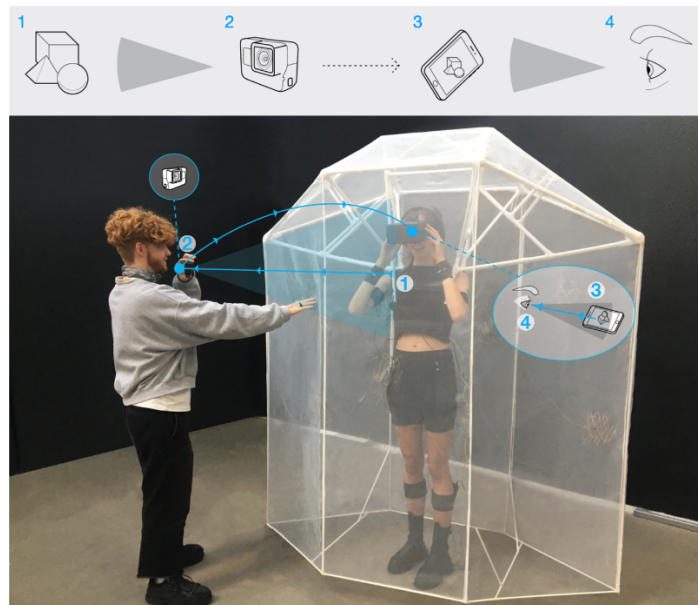


Figure 4: Altering the perception of sight; the camera became a design tool, enabling the participants to design how the person wearing the bodysuit saw the environment by moving the camera, which functioned as ‘technological eyes.’

perspective that I never experienced before” (Participant 1), or “At first, my mind could not comprehend between what I saw and what I felt [...] It also had quite an emotional effect on me, which I did not expect” (Participant 3).

4.2 Changing the Perceived Body-Dress Distinction Using Body-Ownership Illusions

The altering of the perception of sight, sound, and touch that the participants experienced while wearing the bodysuit was described

as also altering their perceived ownership of their own bodies. This attests to the body-ownership illusions created by the bodysuit, which allowed the wearer to experience the world through technological ‘eyes’, ‘ears’, and ‘skin’, which were attached to the outer layer of the bodysuit rather than the wearer’s body.

By placing the camera on the outer layer of the bodysuit, the participants were able to see their environment from different perspectives; these relate to both, their own bodies and non-body related textile surfaces. When the camera was placed on the inside of the outer layer, facing the participant, they were able to see themselves from a third-person perspective, and when placed on the

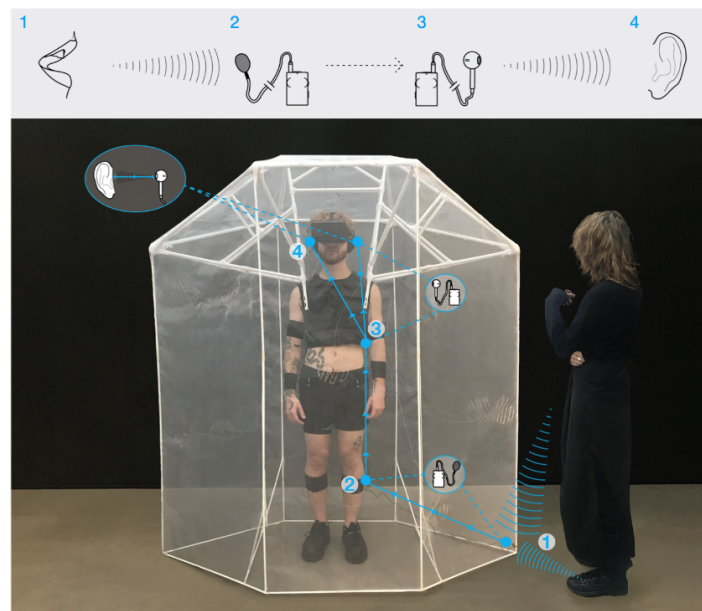


Figure 5: Altering the perception of sound; the microphone and earphones allowed the participants to design how the person wearing the bodysuit heard themselves and others by moving the microphone, which functioned as ‘technological ears.’

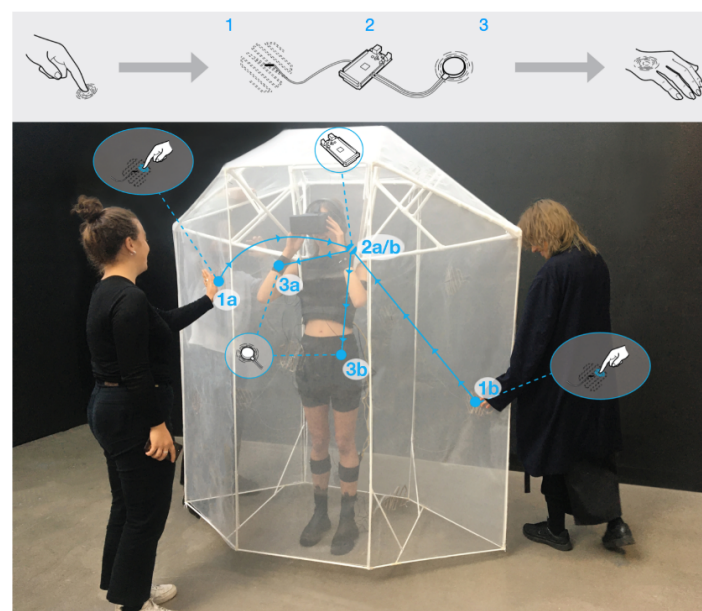


Figure 6: Altering the perception of touch; touching one of the touch sensors embroidered into the outer layer of the bodysuit activated the connected vibration motor attached to the inner layer of the bodysuit, simulating the wearer’s skin being touched. Designing touch was made possible by connecting sensors and actuators in different patterns.

uppermost and outermost points to the left and right of the outer layer the participants were able to see their environment from a non-body-related position. The participants stated that it was important for them that the camera moved and rotated in accordance with the bodysuit to provide them with a sense of direction.

Placing the sound-recording system on the outer layer shifted the perception of hearing to the far left or far right of the participant wearing the bodysuit.

The textile layer of the bodysuit’s outer layer was used as a touch-sensitive ‘skin’. Touch sensors, consisting of conductive yarn, were

embroidered at different points on the outer layer, and activated vibration motors attached to the inner layer. The participants thus felt touch sensations on their skin when the textile of the outer layer was being touched by other participants from the outside, or by themselves from the inside.

Perceiving the environment and oneself through the outer layer of the bodysuit instead of the body itself was described by the participants as an experience that made it difficult for them to distinguish between what they perceived as their body and what was a non-body material (e.g. the bodysuit). This was particularly the case with regard to the sensation of touch. At the beginning, all participants clasped their hands together to provide emotional comfort while wearing the bodysuit, but changed their behavior over time and touched the outer layer of the bodysuit, initiating touch sensations on their body. They felt that this created emotional comfort and helped them to become accustomed to their 'new physical form'.

The participants described the experiences as follows: "My own body became the secondary receiver, and the bodysuit the first, which actually connected me to it" (Participant 1), "In some positions, it felt like the construction was the body and I was just there to watch. That I was just the inner mechanism" (Participant 4).

4.3 Considering the Body and Textiles as Materials of Equal Importance in Fashion Design Processes

The experiences of the participants while wearing the bodysuit suggested the idea of using technology to design different body-dress experiences. While one participant was wearing the bodysuit, the others placed the camera and microphone on different parts of the two layers of the bodysuit and triggered the touch sensors in different patterns. At the beginning, the participants – who functioned as 'perception designers' – altered the perception of sight, sound and touch without putting much thought into how the participant wearing the bodysuit would perceive these stimuli. Instead, they were eager to obscure the perceptive experience of the participant wearing the bodysuit as much as possible.

In the most extreme cases, the camera was positioned by the other participants on one of their bodies so that the person inside the bodysuit saw themselves from the perspective of another person, or held the camera at a distance from the participant wearing the bodysuit in order to change the perception of body ownership to an extreme extent.

A similar approach was taken to sound: the participants not wearing the bodysuit held the microphone close to their own mouths or shoes, so that the person inside the bodysuit perceived others as being closer to them than they really were.

Touch patterns, that confused the person inside the bodysuit so that they could not relate the touch sensation to what they saw, were designed by the other participants.

The effect of this was that the participants felt that they lost their connection to the bodysuit as a perceptive extension when wearing it. They felt uncomfortable, and went back into a state in which they did not move and clasped their hands together. The participants described this experience as follows: "I felt fractured and helpless.

As if my senses were ripped apart and scattered beyond my reach" (Participant 2), or "Felt like I didn't have a body. More like I was isolated somewhere away from everything which was happening in the room and I was just observing someone else being a part of the experiment" (Participant 3).

After each of the participants had worn the bodysuit, however, their behavior changed: the participants who had already worn the bodysuit communicated more directly with participants who had just donned the bodysuit, asking them how they perceived the designed altering of their senses as well as what they experienced as pleasant or unpleasant. The participants increasingly engaged in active conversations with each other, collaboratively designing the alteration of the senses. This shift in designing with the senses improved the experiences of the participants wearing the bodysuit in terms of immersion. The participants described the shift in thinking regarding the senses as follows: "Usually, we tend to disregard the presence of clothes on us since there are a lot of external things happening all at the same time and it usually overpowers the sense/feeling of the layers of the materials being on the body. But this experience made me much more aware of garment and body [...] it further helped to realize how narrow the current way of designing is and how many unexplored ways of approaching fashion design there are" (Participant 2), or "It kind of gave me an understanding of seeing the body in its whole and being more aware of the whole body when designing garments for it" (Participant 4).

5 DISCUSSION

The participants' feedback suggested that the destabilization of their perceived body ownership, which was caused by the drift in bodily perception away from the body towards the layers of the bodysuit, changed their understanding of how fashion designers can consider the body and textiles as materials of equal importance in fashion design processes. The bodysuit's technological components served as design tools for exploring novel artistic expressions in relation to the design of dress outside the limitations of visual features such as colors and silhouettes. The estrangement created by body-ownership illusions encouraged the participants to explore bodily perception as a design feature that can lead to new body-dress experiences. The workshop further suggested that personally experiencing a body-ownership illusion through dress can improve the participants' awareness and understanding of how dress can be designed with the perspective of bodily perception in mind. Before wearing the bodysuit themselves, the participants could not relate to the idea of designing bodily perception using the bodysuit, but this changed after they had experienced its effects. After they had worn the bodysuit, the participants were more considerate and thoughtful when placing the 'technological senses' in relation to the body of the wearer. Moreover, they worked in a more collaborative fashion with the person inside the bodysuit with regard to how and where the 'technological senses' were placed in order to create intriguing and immersive experiences.

The participants' feedback also showed the limitations and challenges that need to be considered and overcome when seeking to improve the experience of designing with sight, sound, and touch through dress. The experienced altering of the senses in general,

which was described by the participants as an immersive experience, could only be achieved to a limited extent due to the fact that the eyes, ears, and skin provide a much more complex sensory apparatus for perceiving the world than is possible to recreate using the technology that was used in the workshop. This was particularly the case with regard to the sense of touch. Touch sensations can include vibrations, pressure, temperature changes, or pain, depending on which receptors in the skin are activated. The vibration motors, however, only provided the sensation of vibration. Similar limitations were mentioned in terms of touch resolution on the skin, given the fact that one feels touch sensations in different intensities depending on the part of the body; for example the hands are highly sensitive to touch while the back is much less so [24]. The participants also felt that the level of immersion varied in quality depending on the degree to which the senses were experienced as being distant from the body. The greater the distance between the body and the 'technological senses', the more helpless the participants inside the bodysuit felt; too great distance caused them to dissociate from the experience. Thus, the experience needs to be designed and technology placed such that experiences relatable to those of the body itself are created.

This study investigated the use of technology for designing dress from a body-related perspective, in contrast to how technology is commonly used in the fashion industry. Exploring the creation of novel and innovative body-dress experiences through the inclusion of technology in fashion design processes is of relevance as dress significantly influences how one perceives one's body and self in everyday life [6, 10, 23, 27]. In this sense, dress can be seen as an everyday practice of re-mixing bodies on a smaller scale, as compared to research practices that combine the body and technology [21, 22, 24, 25]. The increasing digitalization and inclusion of technology in fashion design processes must be considered alongside how one experiences dress when it is worn on the body, rather than focusing solely on making production cycles more sustainable by substituting the body with digital avatars [1, 26]. Limiting the use of technology to increasing the efficiency of production cycles takes away the awareness of how one perceives oneself through the wearing of dress. This study has suggested a wearable design tool that can address this issue. Using the bodysuit to shift perceived body ownership away from one's body and towards the bodysuit increased the participants' awareness of bodily perception as an important design feature when designing dress. In this sense, this study suggests the importance of a holistic approach to designing dress in terms of both the internal and external perspectives.

Future research could further investigate how technologically assisted body-ownership illusions could lead to new body-dress dynamics for the purpose of artistic exploration in fashion. One possible direction could be to further explore body-ownership illusions in relation to dress in the digital space. Rather than using digital avatars as non-perceptive reference material, digital dress could function as perceptive digital body substitutes that could further intensify the experience of dress as a bodily extension. This would contribute to the discussion of body and dress as one in the digital realm from the perspective of fashion design.

6 CLOSING REMARKS

This study investigated novel approaches to including technology in the design process for items of dress, and suggested potential artistic expressions. This study proposes that technology can be used in design processes to raise awareness of the importance of designing dress from and for a body-related external perspective, as well as from and for a body-related internal perspective. The bodysuit that was used in the workshop served as a wearable design tool for altering the perception of sight, sound, and touch, which caused the destabilization of perceived body-dress distinctions. This in turn led to the emergence of an improved understanding of the design of dress among the participants.

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COMMENTARY

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I enjoyed reading about this project and I'm hoping it will foster discussion around the idea of remixed bodies and augmented perception.

What the author calls “perception as design material” is – to me – an instance of remixed bodies. This is something we are seeing more and more of in the HCI community and it's an area I find super exciting. I'm immediately reminded of Jun Nishida's beautiful work which allows us to experience the world from the physical point of view of a child. Just as the vision of participants in the

current work is shifted, Jun's prototype shifts the point of view of a grown person, to see from the perspective of a child [8]. In follow up work, Jun also provides a type of exoskeleton, which allows us to feel things with tiny hands [7], remixing not only input, but also output of the sensorimotor system.

Sang-Won Leigh explores similar ideas, by adding extra limbs to the body [4]. However, he too creates a unique spin on things by exploring the spectrum of devices which extend the body, to devices which collaborate with the body [5]. Here Sang is not only remixing the physical body – which he refers to as changing the “external morphology” – but also the control structure, the “internal morphology” [3].

Yet another exciting take on remixed bodies includes work by Ana Tajadura-Jiménez and Nadia Bianchi-Berthouze. They show that by changing the feedback produced from interactions with the body, the perception of one's own body can be changed [11], for example changing one's perceived weight. They further show that by manipulating the auditory feedback of the body, absurd body-image illusions can be created, such as the sensation of a growing finger [12].

Computer-science has unique opportunities of contributing to this space methodologically. An example of this is work by MacIntosh and colleagues from Copenhagen University, who reformulate body-remixing into an optimization problem. They show how bodies might be ‘improved’ to better perform specific tasks [6]. This space of remixed bodies is broad and exciting, and one of the new areas within HCI which I am most excited about, so I hope the paper at hand will contribute to this discussion.

In the manuscript I reviewed, there was some relatively strong wording regarding the ‘success’ of the sensory mediation. I noted this critically in my initial review, but it actually points towards a larger problem: If we are using perception as a design material how do we evaluate if something is ‘good’? If the goal shifts from the utilitarian goal of representing something so one can grasp the information as fast as possible to ‘good mediation’, how can (or should?) we evaluate it?

I'm reminded of discussions I had with Gershon Dublon who shared with me his frustration of finding a way to evaluate his work on sensory augmentation which adequately captured what he cared about (this was in the context of augmented auditory perception in Tidmarsh [9]). I could relate to this well because of my own early experimentation with telepresence avatars. It was impossibly hard to separate the physical functioning of the device with the experience of satisfactory mediation [10].

Antal Haans explored some ways of sidestepping these problems. Rather than asking participants to introspect on their experience, Antal tried to measure behaviors which would occur had the sensory experience been unmediated. The rational being that if the mediation were ‘good’ it would induce the same types of behaviors an unmediated stimulus would [1]. Sadly, the work – to my best knowledge – had inconclusive results [2] and did not see much follow up. When we try to evaluate such sensory experiences, I wonder if the HCI community should spend more time looking at what other areas in research have established. I wonder if research in psychology which looks at full-body illusions, body-ownership, or proprioceptive drift might not provide useful frameworks for better evaluating such systems?

In summary, I'm hoping that this paper opens up discussions around these topics yet again. Maybe such discussions will help us better understand the commonalities in terms of underlying mechanisms of existing research in HCI. I'd hope that as computer-scientists we can provide a unique point of view and unique methodological approaches to complement existing research in this area in psychology. And to do this, many of us (including myself) have some homework to do, so as to not reinvent the wheel.

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